MULUNGUSHI UNIVERSITY

DEPARTMENT OF COMPUTER SCIENCE

ICT261: OBJECT-ORIENTED PROGRAMMING II — UNIT 4 LAB REPORT

STUDENT INFORMATION:

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Student Number: 202510599

Program: Computer Science

Course Code: ICT261 — Unit 4 (Java Exception Handling and Error Management)

Date: November 2, 2026

ACTIVITY A: EXCEPTION RISK ASSESSMENT & PREDICTION

During automated stream reading and validation of text-based student record data ("marks.txt"), five distinct exception conditions may arise:

1. FileNotFoundException:

   Occurs when the runtime environment fails to locate the targeted file due to an invalid path name or an absent file in the working directory.

2. IOException:

   Raised during hardware or stream read interruptions, unexpected file stream terminations, or insufficient filesystem permissions.

3. NumberFormatException:

   Generated when non-numeric text strings (such as letters or invalid symbols)

are passed to Integer.parseInt() during mark conversion.

4. IllegalArgumentException:

Thrown by domain validation logic when a line fails layout constraints (not exactly two fields) or when numerical marks violate boundary limits (< 0 or > 100).

5. DuplicateStudentException:

A custom application runtime exception designed to detect and prevent

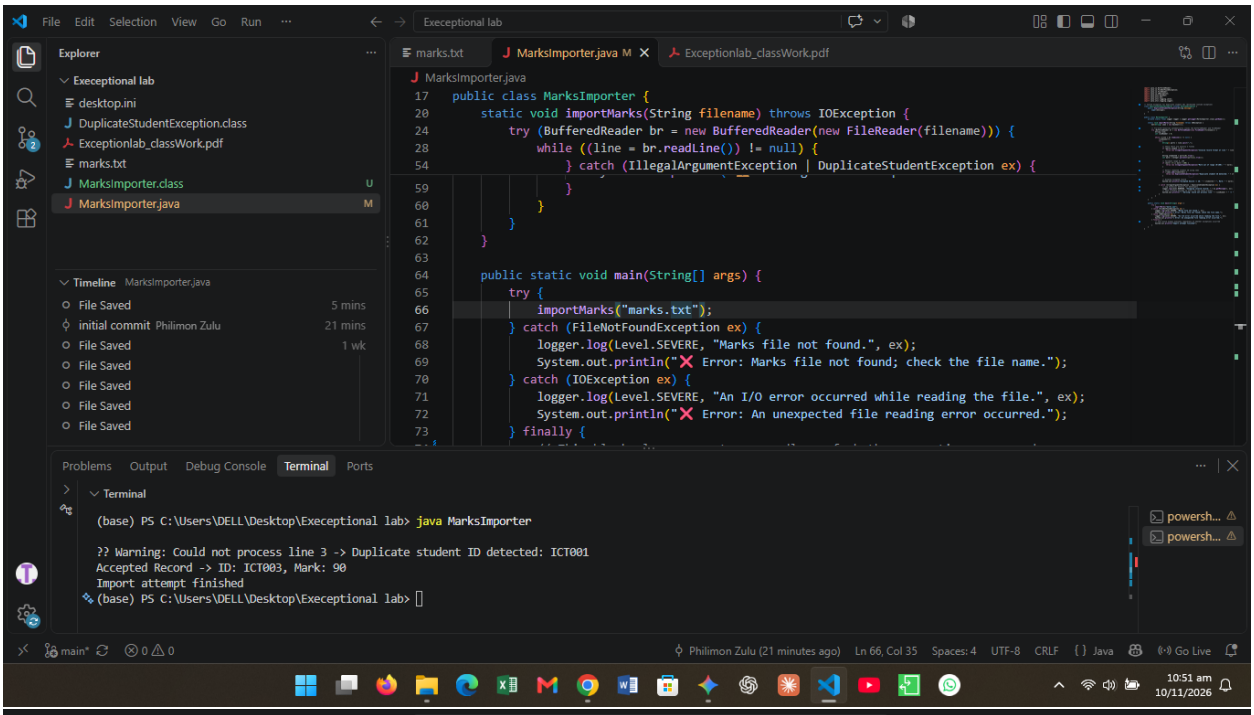
ACTIVITY D: SOFTWARE ENGINEERING REVIEW & DESIGN STRATEGY

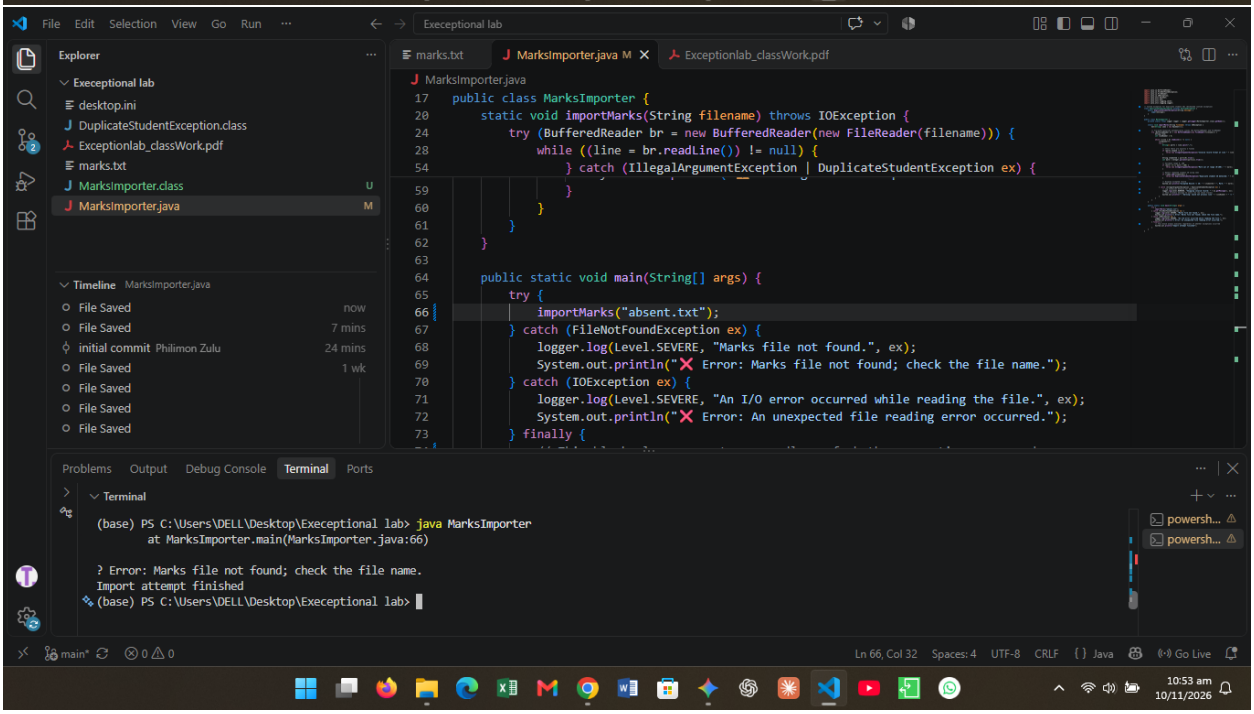
1. Architectural Rationale for Partial Processing:

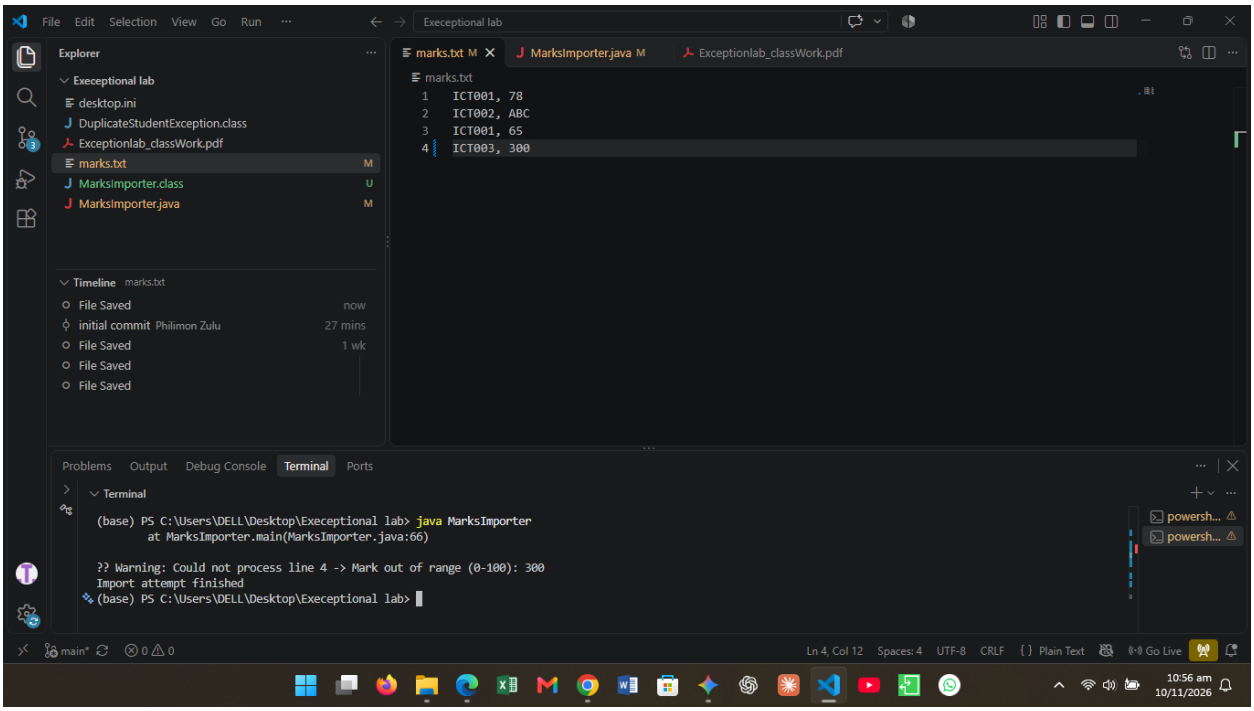
In production grade software, whole-batch abortion due to localized formatting flaws creates severe operational bottlenecks. Implementing per-record exception handling creates a resilient system that isolates corrupted rows, writes diagnostic stack traces to developer logs, informs the operator with user-friendly warnings, and continues importing valid student data without interruption.

2. Evaluation Summary for Test Cases (T1–T5):

- T1 (Standard Valid Entry): Parsed correctly formatted student IDs and accepted valid numerical marks.
- T2 (Malformed Data Type): Intercepted NumberFormatException, logged the line fault, and skipped row execution.
- T3 (Out-of-Bounds Grade): Triggered IllegalArgumentException for marks failing range checks (0–100).
- T4 (Redundant ID Record): Enforced unique tracking via HashSet and threw DuplicateStudentException.
- T5 (Non-Existent Source File): Caught FileNotFoundException cleanly and ensured execution of the 'finally' block.


File Edit Selection View Go Run
Exceptional lab
Explorer
Exceptional lab
desktop.ini
DuplicateStudentException.class
ExceptionIab_classWork.pdf
marks.txt
MarksImporter.class U
MarksImporter.java M
Timeline MarksImporter.java
File Saved 5 mins
initial commit Philimon Zulu 21 mins
File Saved 1 wk
File Saved
File Saved
File Saved
File Saved
marks.txt
MarksImporter.java M
ExceptionIab_classWork.pdf
MarksImporter.java
17 public class MarksImporter {
20 static void importMarks(String filename) throws IOException {
24 try (BufferedReader br = new BufferedReader(new FileReader(filename))) {
28 while ((line = br.readLine()) != null) {
54 } catch (IllegalArgumentException | DuplicateStudentException ex) {
59 }
60 }
61 }
62 }
64 public static void main(String[] args) {
65 try {
66 importMarks("marks.txt");
67 } catch (FileNotFoundException ex) {
68 logger.log(Level.SEVERE, "Marks file not found.", ex);
69 System.out.println("❌ Error: Marks file not found; check the file name.");
70 } catch (IOException ex) {
71 logger.log(Level.SEVERE, "An I/O error occurred while reading the file.", ex);
72 System.out.println("❌ Error: An unexpected file reading error occurred.");
73 } finally {
Problems Output Debug Console Terminal Ports
Terminal
(base) PS C:\Users\DELL\Desktop\Exceptional lab> java MarksImporter
?? Warning: Could not process line 3 -> Duplicate student ID detected: ICT001
Accepted Record -> ID: ICT003, Mark: 90
Import attempt finished
(base) PS C:\Users\DELL\Desktop\Exceptional lab>
powersh...
powersh...
main* 0 0
Philimon Zulu (21 minutes ago) Ln 66, Col 35 Spaces: 4 UTF-8 CRLF { } Java Go Live
10:51 am
10/11/2026


File Edit Selection View Go Run
Exceptional lab
Explorer
Exceptional lab
desktop.ini
DuplicateStudentException.class
ExceptionIab_classWork.pdf
marks.txt
MarksImporter.class U
MarksImporter.java M
Timeline MarksImporter.java
File Saved now
File Saved 7 mins
initial commit Philimon Zulu 24 mins
File Saved 1 wk
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marks.txt
MarksImporter.java M
ExceptionIab_classWork.pdf
MarksImporter.java
17 public class MarksImporter {
20 static void importMarks(String filename) throws IOException {
24 try (BufferedReader br = new BufferedReader(new FileReader(filename))) {
28 while ((line = br.readLine()) != null) {
54 } catch (IllegalArgumentException | DuplicateStudentException ex) {
59 }
60 }
61 }
62 }
64 public static void main(String[] args) {
65 try {
66 importMarks("absent.txt");
67 } catch (FileNotFoundException ex) {
68 logger.log(Level.SEVERE, "Marks file not found.", ex);
69 System.out.println("❌ Error: Marks file not found; check the file name.");
70 } catch (IOException ex) {
71 logger.log(Level.SEVERE, "An I/O error occurred while reading the file.", ex);
72 System.out.println("❌ Error: An unexpected file reading error occurred.");
73 } finally {
Problems Output Debug Console Terminal Ports
Terminal
(base) PS C:\Users\DELL\Desktop\Exceptional lab> java MarksImporter
at MarksImporter.main(MarksImporter.java:66)
? Error: Marks file not found; check the file name.
Import attempt finished
(base) PS C:\Users\DELL\Desktop\Exceptional lab>
powersh...
powersh...
main* 0 0
Ln 66, Col 32 Spaces: 4 UTF-8 CRLF { } Java Go Live
10:53 am
10/11/2026


File Edit Selection View Go Run
Exceptional lab
Explorer
Exceptional lab
desktop.ini
DuplicateStudentException.class
Exceptionlab_classWork.pdf
marks.txt M
MarksImporter.class U
MarksImporter.java M
Timeline marks.txt
File Saved now
initial commit Philimon Zulu 27 mins
File Saved 1 wk
File Saved
File Saved
marks.txt M
MarksImporter.java M
Exceptionlab_classWork.pdf
marks.txt
1 ICT001, 78
2 ICT002, ABC
3 ICT001, 65
4 ICT003, 300
Problems Output Debug Console Terminal Ports
Terminal
(base) PS C:\Users\DELL\Desktop\Exceptional lab> java MarksImporter
at MarksImporter.main(MarksImporter.java:66)
?? Warning: Could not process line 4 -> Mark out of range (0-100): 300
Import attempt finished
(base) PS C:\Users\DELL\Desktop\Exceptional lab>
powersh...
powersh...
main*
Ln 4, Col 12 Spaces: 4 UTF-8 CRLF Plain Text Go Live
10:56 am
10/11/2026